

Introduction to Uniform Centripetal Motion

Name: _____

Date: _____

Part A: Reminder — Linear Motion (Straight-Line)

What do we call something that has both magnitude and direction?

What are some examples of vectors?

What are some examples of scalars?

Fill in: Circle the correct choice.

- 1) Speed is a (**scalar** / **vector**)
- 2) Velocity is a (**scalar** / **vector**)
- 3) Force is a (**scalar** / **vector**)
- 4) Mass is a (**scalar** / **vector**)

In straight-line motion, if an object moves with **constant velocity**, then its acceleration is:

Why is this the case?

Key idea: In **linear motion**, constant velocity $\Rightarrow$ zero acceleration.

Part B: Why Circular Motion Is Different

For objects moving in a **circle**, it is possible to have constant speed but non-zero acceleration.

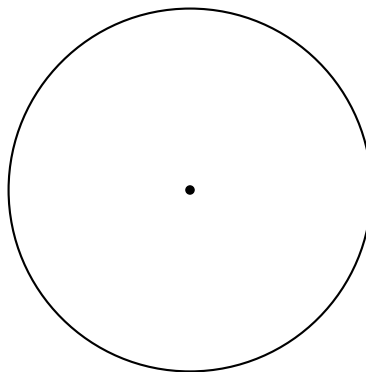
Why is that?

Let's visualize this.

Key idea: Any change in velocity (speed OR direction) means acceleration is not zero.

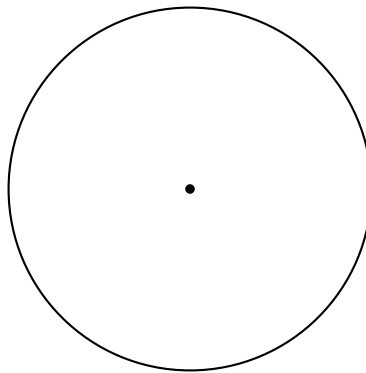
Part C: Circle Geometry Refresher

What are the elements of a circle?



Part D: Concept Questions & Velocity and Acceleration in Uniform Centripetal Motion

Suppose you are in a car that makes a sharp left turn, do we go left or right in the car?



In uniform circular motion:

- The **velocity** $\vec{v}$ is always **tangent** to the circle (points along the direction of motion).
- The **acceleration** $\vec{a}_c$ is always **toward the center** (centripetal = “center-seeking”).

Part E: Real-World Examples

List a few real-world examples of uniform centripetal motion (constant speed in a circle).

1) _____

2) _____

3) _____

Part F: Centripetal Acceleration and Centripetal Force

What is the equation for centripetal acceleration?

What is the equation for centripetal force?

In what ways can the force be represented?

Check your understanding:

- a) If speed v doubles, what happens to a_c ?

- b) If radius r doubles (same v), what happens to a_c ?

- c) If mass m doubles (same v), what happens to F_c ?

Practice Problems

1. A student swings a 0.50 kg rubber ball in a horizontal circle using a string. The ball moves at a constant speed of $v = 4.0 \text{ m/s}$ in a circle of radius $r = 1.6 \text{ m}$.

- a) Calculate the centripetal acceleration.
- b) What is the centripetal force on the ball?
- c) What provides this centripetal force in this situation?

2. A 0.50 kg puck slides at a constant speed of $v = 4.0 \text{ m/s}$ on a horizontal circular track of radius $r = 1.6 \text{ m}$. The puck does not slip as it moves in a circle.

- a) Calculate the centripetal acceleration.
- b) What is the centripetal force on the puck?
- c) What provides this centripetal force in this situation?

3. A car goes around a circular track of radius $r = 50$ m. The maximum centripetal acceleration the tires can handle is $a_c = 6.0 \text{ m/s}^2$ before slipping.

a) What is the maximum speed the car can have without slipping?

b) If the radius were **smaller**, would the safe maximum speed increase or decrease? Explain briefly.

4. A 1200 kg car travels around a curve of radius $r = 80$ m at a constant speed of $v = 18$ m/s.

a) Find the centripetal acceleration.

b) Find the required centripetal force.

c) Compare this centripetal force to the car's weight mg . Is F_c larger or smaller than mg ?